

4.1 Embodied Cognition

Andy Clark, Barbara Webb — perception, action, behaviour
↳ philosophy of cog. sciences → embodied cognition

Group interested in minimal minds

eg: game dynamic
walking machines
(Cornell Univ.)

how much / little brain is needed to
make a robot do intelligent things /
explain adaptive behaviour

↓
no motor or controllers, motion begins by falling forward on a
sliding slope

↓
understanding and building the physical system tells us more about how
we walk than trying to introspectively think the process

perception of the world

we also perceive what we see — sensory input — we have feedback
have intentions

are we
more than
machines?

E.g.

→ Desert Ants → their behaviour is interesting because they don't follow
chemical trails but can navigate through complex environments

↓
Up until now it was thought that ants could should recall their
position along the route

↓
but a robot can mimic their behaviour without recalling the
position, instead → alters zig-zig proportionally to how unfamiliar
its current heading appears → if familiar → continue
if " " → swing around until seeing
something familiar

other adaptations → a to, part of its eye → sensitive to polarized light

Physical Structures are designed to solve problems —

← can deduce position of sun even when sun can't be seen.

4.2

→ Also consider reliable features of the environment

↓
Bluefin tuna

↓

making short-term changes to its immediate environment to improve swimming to get back energy from environment to work better

→ use the currents to move
→ flip its tail to create small vortices of water → move more efficiently

↓

new generation of swimmers

↑
odor than a gradient

ANTS

bury eggs → have odor → more ants bury more eggs
ants build a wall (made of particles) following 2 rules:

- each ant will pick up a particle and if it carries the particle into the nest — then carries nest entrance and lay particle inside (which following gradient)
- other ants will lay the particles where the 1st particle is

How does this scale up to cognition?

Embodied interaction simplified learning → e.g. Babybot work
(Giorgio Metta, Genova)
moves limbs around and catalogues the
patterns of interaction it experiences

James Gibson → 1950s → argued what we understand by a
chair is the concept of sitability

E.g. automated cars / robots directing themselves

→ needs to be grounded and not too abstract

→ don't need to know where every possible obstacle is

↓
but an abstract map of where we want to go
sensory systems that
- keeps us on the road
- allow us to take immediate action if
we see a funny object

!!! Avoid the "Naked Brain fallacy" → fallacy of assuming all
the interesting cognitive action is always going on in the brain

↓ success: multilayer control system

→ lowest level of control involved in interaction
with the world

→ leaves immediate processes
to work on their own

↓ frees space for

→ higher / conceptual level → how to get from
A to B

↓
long-term planning